Second order linear differential equations:

A general second order linear ordinary differential equation (ODE) is of the form

$$a_2(x)\,y'' + a_1(x)\,y' + a_0(x)\,y = b(x).$$

The standard form of a second order linear ODE is given by $y'' + p(x)\,y' + q(x)\,y = r(x)$. — (1)

Remark: There is no formula to find all solutions of such an ODE, we study the existence, uniqueness and no of solutions of such ODE's.

If $r(x) = 0$ in Eq. (1), then the ODE is said to homogeneous. Otherwise, it is called non-homogeneous.

Initial value problem - Existence & Uniqueness:

An initial value problem of a second order homogeneous linear ODE is of the form

$$y'' + p(x)\,y' + q(x)\,y = 0, \quad y(x_0) = a, \quad y'(x_0) = b,$$

where $p(x)$ and $q(x)$ are continuous on an open interval I with $x_0 \in I$. It has a unique solution $y(x)$ in the interval I.

Linearly independent functions:

The functions $y_1(x)$ and $y_2(x)$ are said to be linearly independent (LI) on an open interval I if

$c_1 y_1(x) + c_2 y_2(x) = 0$ for all $x \in I$ and constants c_1 & c_2, then $c_1 = c_2 = 0$. Otherwise, it is called linearly dependent.

Example 1: $y_1(x) = e^{m_1 x}$ and $y_2(x) = e^{m_2 x}$, $m_1 \neq m_2$, are LI on $\mathbb{R}$. Let $c_1 e^{m_1 x} + c_2 e^{m_2 x} = 0$. Put $x = 0$, this gives $c_1 + c_2 = 0$. Now, differentiate w.r.t. x,

$c_1 m_1 e^{m_1 x} + c_2 m_2 e^{m_2 x} = 0$. Put $x = 0$, this gives,

$c_1 m_1 + c_2 m_2 = 0$. By solving $c_1 + c_2 = 0$ & $m_1 c_1 + m_2 c_2 = 0$, we get $c_1 = c_2 = 0$ $(\because m_1 \neq m_2)$.

Example 2: $y_1(x) = \cos ax$ and $y_2(x) = \sin ax$, $a \neq 0$, are LI on $\mathbb{R}$.

Wronskian: The wronksian of two differential functions $y_1(x)$ and $y_2(x)$ is defined by

$$W(y_1, y_2)(x) := \det \begin{bmatrix} y_1(x) & y_2(x) \\ y_1'(x) & y_2'(x) \end{bmatrix}.$$

For simplicity, we denote the Wronksian by $W(x)$ or W.

Test for LI: Suppose $y''(x) + p(x)y' + q(x)y = 0$ — (2) has continuous coefficients on an open interval I. Then

(i) two solutions y_1 and y_2 of ODE (2) on I are LD iff their Wronksian is 0 at some $x_0 \in I$.

(ii) Wronksian $= 0$ for some $x = x_0$ implies $W = 0$ on I.

(iii) If there exists $x_1 \in I$ at which $W \neq 0$, then y_1 & y_2 are LI on I.

Proof: (i) Let y_1 and y_2 be LD. Then $y_2 = cy_1$ for some constant c. Then $W(y_1, y_2) = W(cy_1, cy_1) = 0$ (by the definition).

Conversely, suppose $W(y_1, y_2) = 0$ for some $x_0 \in I$, i.e.,

$$W(x_0) = \begin{vmatrix} y_1(x_0) & y_2(x_0) \\ y_1'(x_0) & y_2'(x_0) \end{vmatrix} = 0.$$

Then the system of linear equations

$$\begin{bmatrix} y_1(x) & y_2(x_0) \\ y_1'(x) & y_2'(x_0) \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \text{— (3)}$$

has non zero solution (as the determinant of the coefficient matrix is zero). Let $[c_1, c_2]^T$ be a non zero solution of (3) and $y(x) = c_1 y_1(x) + c_2 y_2(x)$. We have $y(x_0) = y'(x_0) = 0$ (by Eq. (3)). Note that $y(x) = 0$ is a solution of Eq. (2). By existence & uniqueness of IVP (see above), $y(x) = 0$ is the unique solution of Eq. (2) with $y(x_0) = y'(x_0) = 0$. Thus $y(x) = c_1 y_1(x) + c_2 y_2(x) = 0$ with c_1 and c_2 both not identically zero. Hence, y_1 and y_2 are LD.

(ii) Suppose $W = 0$ for some $x = x_0$, then by (i), y_1 & y_2 are LD this implies $W(x) = 0$ for all $x \in I$.

(iii) Suppose $W(x_1) \neq 0 \Rightarrow y_1$ & y_2 can not be LD $\Rightarrow$ y_1 & y_2 are LI. $\square$

Example 3: Verify that $y_1 = e^{2x}$ and $y_2 = e^{3x}$ are solutions of $y'' - 5y' + 6y = 0$. Apply the above test to see that y_1 and y_2 are LI.

Example 4: Verify that $y_1 = \cos x$ and $y_2 = \sin x$ are solutions of $y'' + y = 0$. Apply the above test to see that y_1 and y_2 are LI.

Remark (i) The continuity of $p(x)$ and $q(x)$ is needed in the above test, e.g., take $y'' + p(x)y' + q(x)y = 0$, where

$$p(x) = \begin{cases} -\frac{6}{x}, & x \neq 0, \\ 0, & x = 0 \end{cases} \quad \text{and} \quad q(x) = \begin{cases} \frac{12}{x^2}, & x \neq 0, \\ 0, & x = 0. \end{cases}$$

Here, $y_1 = x^3$ and $y_2 = x^4$ are LI solutions but $W(0) = 0$.

(ii) y_1 and y_2 have to be solutions of Eq. (2). Otherwise, the test is not true. E.g., take $y_1 = x^2$ and

$$y_2 = \begin{cases} x^2 & \text{if } x \geq 0 \\ -x^2 & \text{if } x < 0, \end{cases}$$ then $W(x) = 0$ for all $x \in \mathbb{R}$, but

y_1 and y_2 are LI.

Theorem: If y_1 and y_2 are solution of a homogeneous linear ODE (Eq. (2)) on an open interval I then $c_1y_1 + c_2y_2$, c_1 & c_2 are constants, is again a solution of Eq. (2).

Example 5: In Example 3, show that $y(x) = c_1e^{2x} + c_2e^{3x}$ is a solution of $y'' - 5y' + 6y = 0$.

Example 6: In Example 4, show that $y(x) = c_1\cos x + c_2\sin x$ is a solution of $y'' + y = 0$.

Example 7: Show that $y(x) = c_1x^2 + c_2x^3$ is a solution of $y'' - \frac{4}{x}y' + \frac{6}{x^2}y = 0$.

Remark: The above theorem is not true for non-homogeneous linear ODE. E.g., take $y'' + y = 1$. Here $y_1 = 1 + \cos x$ and $y_2 = 1 + \sin x$ are solutions but $y_1 + y_2$, $2y_1$ & $3y_2$ are not solutions.

Definition: If y_1 and y_2 are LI solutions of Eq.(2) on an open interval I then $\{y_1, y_2\}$ is called a basis of Eq.(2) on I and $y(x) = c_1 y_1(x) + c_2 y_2(x)$ is called a general solution of Eq.(2) on I.

In Examples 5, 6 & 7 the given solutions are general solutions of respective ODE.

Result: If $p(x)$ and $q(x)$ are continuous on an open interval I, then Eq.(2): $y'' + p(x)y' + q(x)y = 0$ has a basis of solutions on I.

Proof: Consider IVP's

$$y'' + p(x)y' + q(x)y = 0, \quad y(x_0) = 1, \quad y'(x_0) = 0;$$

$$y'' + p(x)y' + q(x)y = 0, \quad y(x_0) = 0, \quad y'(x_0) = 1.$$

By existence & uniqueness of IVP, the above problem have unique solutions y_1 and y_2 respectively on I. Now,

$$W(x_0) = \begin{vmatrix} y_1(x_0) & y_2(x_0) \\ y_1'(x_0) & y_2'(x_0) \end{vmatrix} = \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1 \neq 0.$$

By the above test, y_1 and y_2 are LI. ($\because y_1$ & y_2 are solution of Eq(2) as well). Hence, they form a basis of Eq.(2).

Exercise: If $\{y_1, y_2\}$ is a basis of Eq(2) on I with $p(x)$ and $q(x)$ continuous on I, then show that every solution y of Eq.(2) has the form $y = c_1 y_1 + c_2 y_2$, where c_1 and c_2 are constants.

Hint: Write $y = c_1 y_1 + c_2 y_2$ and find c_2 & c_2 by forming a system of linear eqs $y(x_0) = c_1 y_1(x_0) + c_2 y_2(x_0)$ and $y'(x_0) = c_1 y_1'(x_0) + c_2 y_2'(x_0)$. Apply Existence & Uniqueness of IVP on $u(x) = y - c_1 y_1 - c_2 y_2$, $u(x_0) = 0 = u'(x_0)$.

Why we have to apply existence & uniqueness theorem?
